18. If A is 3×3 matrix and $|A| = 4$ then $|A^{-1}|$ is equal to

(a) $\frac{1}{4}$ (b) $\frac{1}{16}$ (c) 2 (d) 4

19. If A is a square matrix of order 3 and $|A| = 3$ then $|adjA|$ is equal to

(a) 81 (b) 27 (c) 3 (d) 9

20. The value of $\begin{vmatrix} x & x^2 - yz & 1 \\ y & y^2 - zx & 1 \\ z & z^2 - xy & 1 \end{vmatrix}$ is

(a) 1 (b) 0

(c) –1 (d) $-xyz$

21. If $A = \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix}$, then $|2A|$ is equal to

(a) $4\cos 2\theta$ (b) 4

(c) 2 (d) 1

22. If $\Delta = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$ and A_{ij} is cofactor of a_{ij}, then value of Δ is given by

(a) $a_{11}A_{31} + a_{12}A_{32} + a_{13}A_{33}$

(b) $a_{11}A_{11} + a_{12}A_{21} + a_{13}A_{31}$

(c) $a_{21}A_{11} + a_{22}A_{12} + a_{23}A_{13}$

(d) $a_{11}A_{11} + a_{21}A_{21} + a_{31}A_{31}$

23. If $\begin{vmatrix} x & 2 \\ 8 & 5 \end{vmatrix} = 0$ then the value of x is

(a) $\frac{-5}{6}$ (b) $\frac{5}{6}$

(c) $\frac{-16}{5}$ (d) $\frac{16}{5}$

24. If $\begin{vmatrix} 4 & 3 \\ 3 & 1 \end{vmatrix} = -5$ then the value of $\begin{vmatrix} 20 & 15 \\ 15 & 5 \end{vmatrix}$ is

(a) –5 (b) –125

(c) –25 (d) 0

25. If any three rows or columns of a determinant are identical then the value of the determinant is

(a) 0 (b) 2 (c) 1 (d) 3

Miscellaneous Problems

1. Solve: $\begin{vmatrix} x & 2 & -1 \\ 2 & 5 & x \\ -1 & 2 & x \end{vmatrix} = 0$

2. Evaluate $\begin{vmatrix} 10041 & 10042 & 10043 \\ 10045 & 10046 & 10047 \\ 10049 & 10050 & 10051 \end{vmatrix}$

3. Without actual expansion show that the value of the determinant $\begin{vmatrix} 5 & 5^2 & 5^3 \\ 5^2 & 5^3 & 5^4 \\ 5^4 & 5^5 & 5^6 \end{vmatrix}$ is zero.

4. Show that $\begin{vmatrix} 0 & ab^2 & ac^2 \\ a^2b & 0 & bc^2 \\ a^2c & b^2c & 0 \end{vmatrix} = 2a^3b^3c^3$.

5. If $A = \begin{bmatrix} 1 & 1 & 1 \\ 3 & 4 & 7 \\ 1 & -1 & 1 \end{bmatrix}$ verify that $A(adjA) = (adjA)(A) = |A|I_3$.

6. If $A = \begin{bmatrix} 3 & -1 & 1 \\ -15 & 6 & -5 \\ 5 & -2 & 2 \end{bmatrix}$ then, find the Inverse of A.

7. If $A = \begin{bmatrix} 1 & -1 \\ 2 & 3 \end{bmatrix}$ show that $A^2 - 4A + 5I_2 = O$ and also find A^{-1}

8. Solve by using matrix inversion method: $x - y + z = 2, 2x - y = 0, 2y - z = 1$.

9. The cost of 2 Kg of Wheat and 1 Kg of Sugar is ₹70. The cost of 1 Kg of Wheat and 1 Kg of Rice is ₹70. The

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